

Course Design of Deep Learning and Computer Vision

人工智能1班、2班
AI-1 & AI-2

Project

- The practical course offers two optional topics, and students may choose either one. Students should work in groups of 3-4.
- Deadline: Week 18. (7/3)

Deliverables:

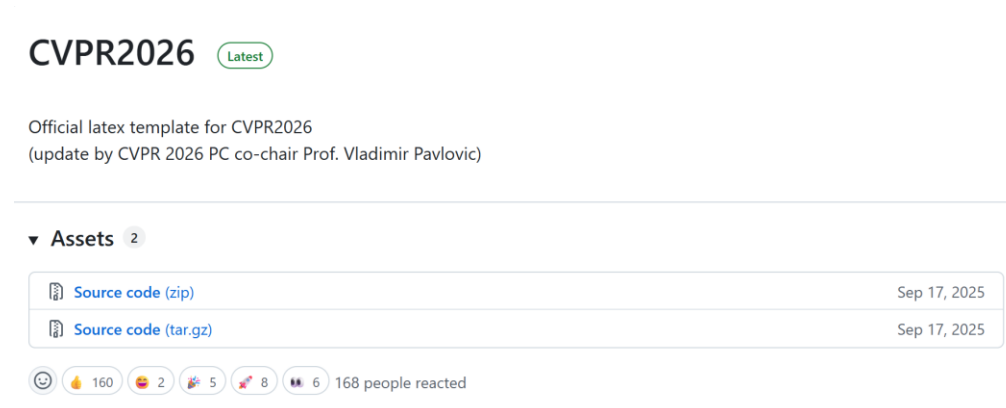
1. Final Technical Report (English, CVPR format)

Includes architecture, methods, implementation, experiments, limitations, etc.

<https://github.com/cvpr-org/author-kit/releases>

2. Code Repository + README

1. Include installation steps, dependencies, and run instructions
2. Include the test cases and the corresponding results.



Technical Report Requirements

Your report (≤ 8 pages including figures and tables, excluding references; appendix (optional) unlimited; English; CVPR template) must include:

- ✓ **Abstract:** Briefly describe your problem, approach, and key results. Should be no more than 300 words.
- ✓ **Introduction:** Describe the problem you are working on, why it's important, and an overview of your results
- ✓ **Related Work:** Discuss published work that relates to your project. How is your approach similar or different from others?
- ✓ **Methods:** Discuss your approach for solving the problems that you set up in the introduction. Why is your approach the right thing to do? Did you consider alternative approaches? You should demonstrate that you have applied ideas and skills built up during the quarter to tackling your problem of choice. It may be helpful to include figures, diagrams, or tables to describe your method or compare it with other methods.
- ✓ **Experiments:** Describe the data you are working with for your project. What type of data is it? Where did it come from? How much data are you working with? Did you have to do any preprocessing, filtering, or other special treatment to use this data in your project? Discuss the experiments that you performed to demonstrate that your approach solves the problem. The exact experiments will vary depending on the project, but you might compare with previously published methods, perform an ablation study to determine the impact of various components of your system, experiment with different hyperparameters or architectural choices, use visualization techniques to gain insight into how your model works, discuss common failure modes of your model, etc. You should include graphs, tables, or other figures to illustrate your experimental results.

Technical Report Requirements

Your report (≤ 8 pages including figures and tables, excluding references; appendix (optional) unlimited; English; CVPR template) must include:

- ✓ **Conclusion:** Summarize your key results - what have you learned? Suggest ideas for future extensions or new applications of your ideas.
- ✓ **Is your paper clearly written and nicely formatted?**
- ✓ **Important:** Include a short statement describing how AI tools were used during your writing/coding.

Overview

- The Course Project is an opportunity for you to apply what you have learned in class to a problem of your interest. Potential projects usually fall into these two tracks:
- **Applications:** If you're coming to the class with a specific background and interests, we'd love to see you apply vision models learned in this class to problems related to your particular domain of interest. Pick a real-world problem and apply computer vision models to solve it.
- For applications, this type of projects would include the careful data preparation, an appropriate loss function, details of training and cross-validation and good test set evaluations and model comparisons.
- For example, ConvNets also run in real time on mobile phones - building an interesting mobile application could be a good project.

Overview

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- **Models:** You can build a new model (algorithm) or a new variant of existing models, and apply it to tackle vision tasks. This track might be more challenging, and sometimes leads to a piece of publishable work.
- One restriction to note is that this is a Computer Vision class, so your project should involve pixels of visual data in some form somewhere. E.g. a pure NLP project is not a good choice, even if your approach involves ConvNets/Transformers.
- Please find a **baseline method/paper** for applications/models.

Ideas

- To inspire ideas, you might also look at recent deep learning publications from top-tier conferences, as well as other resources below.
- **CVPR**: IEEE Conference on Computer Vision and Pattern Recognition
- **ICCV**: International Conference on Computer Vision
- **ECCV**: European Conference on Computer Vision
- **NeurIPS**: Neural Information Processing Systems
- **ICLR**: International Conference on Learning Representations
- **ICML**: International Conference on Machine Learning
- **Kaggle challenges**: An online machine learning competition website. For example, a Yelp classification challenge.

Datasets

- You might also gain inspiration by taking a look at some popular computer vision datasets:
- [ImageNet](#): a large-scale image dataset for visual recognition
- Vision-language datasets: [Visual Genome](#), [Flickr30k](#), [VQA v2](#), [ADE20K](#), [LAION](#)
- [SA-1B](#): dataset of a large number of images and segmentation masks to segment objects in those images
- [COCO](#): large-scale object detection, segmentation, and captioning dataset
- [Open Images](#): a dataset of ~9M images annotated with image-level labels, object bounding boxes, object segmentation masks, visual relationships, and localized narratives
- [Cityscapes Dataset](#): This focuses on semantic understanding of urban street scenes, with pixel-level annotations for various object classes such as cars, pedestrians, and roads
- [DeepFashion](#): a large-scale clothes dataset containing over 800,000 diverse fashion images annotated with bounding boxes, clothing categories, and attributes
- [Hugging face datasets](#): collection of generic datasets available on hugging face
- [Objaverse](#): a large-scale 3D asset database
- [Moments in Time](#): A dataset of one million 3-second videos

Datasets

- [SUN Database](#): a benchmark for scene recognition and object detection with annotated scene categories and segmented objects
- [Places Database](#): a scene-centric database with 205 scene categories and 2.5 millions of labelled images
- [NYU Depth Dataset v2](#): a RGB-D dataset of segmented indoor scenes
- [Microsoft COCO](#): a new benchmark for image recognition, segmentation and captioning
- [Flickr100M](#): 100 million creative commons Flickr images
- [Labeled Faces in the Wild](#): a dataset of 13,000 labeled face photographs
- [Human Pose Dataset](#): a benchmark for articulated human pose estimation
- [YouTube Faces DB](#): a face video dataset for unconstrained face recognition in videos
- [UCF101](#): an action recognition data set of realistic action videos with 101 action categories
- [HMDB-51](#): a large human motion dataset of 51 action classes
- [ActivityNet](#): A large-scale video dataset for human activity understanding
- Others.....

Examples

- Enhancing Segment Anything Model (SAM) for Brain Tumor Image Segmentation
https://cs231n.stanford.edu/2025/papers/text_file_840592918-Final%20Report%20Submission.pdf
- WEBSIGHT: A Vision-First Architecture for Robust Web Agents
https://cs231n.stanford.edu/2025/papers/text_file_840592471-CS_231N_Final_Report.pdf
- Bridging Vision and Language for Sign Language Understanding
https://cs231n.stanford.edu/2025/papers/text_file_840485531-CS231N_Final_Report.pdf
- Teaching Deep Convolutional Neural Networks to Play Go
<https://arxiv.org/pdf/1412.3409>